Questions

Q1.

* Figure 16 illustrates an effect that can be explained using the ideas of pressure, force and area.

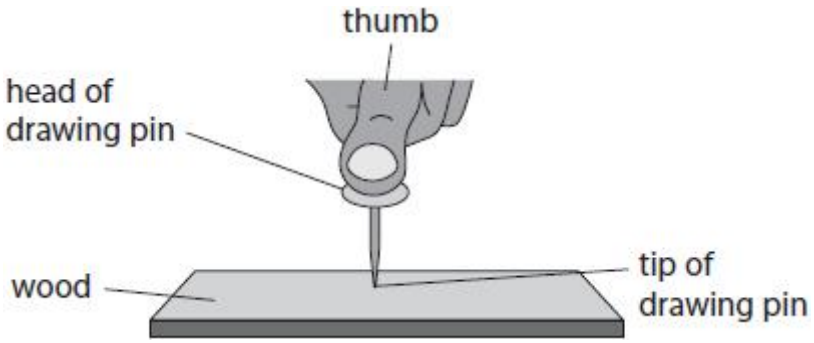

Figure 16

Explain why the tip of the drawing pin goes into the wood but the head of the drawing pin does not go into the thumb.

(6)

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(Total for question = 6 marks)

Q2.

A swimming pool contains fresh water.

The pressure 2 m below the surface of the swimming pool is measured.

The pressure 10 m below the surface of the sea is also measured.

The pressure 10 m below the surface of the sea is greater than the pressure 2 m below the surface of the swimming pool.

State **two** reasons why the pressure is greater 10 m below the surface of the sea.

(2)

1 .....

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2 .....

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(Total for question = 2 marks)

Q3.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A student carries out an investigation to show how pressure varies with depth in water.

A pressure sensor is attached to a rule.

The rule and pressure sensor are lowered into the water in a tank, as shown in Figure 25.

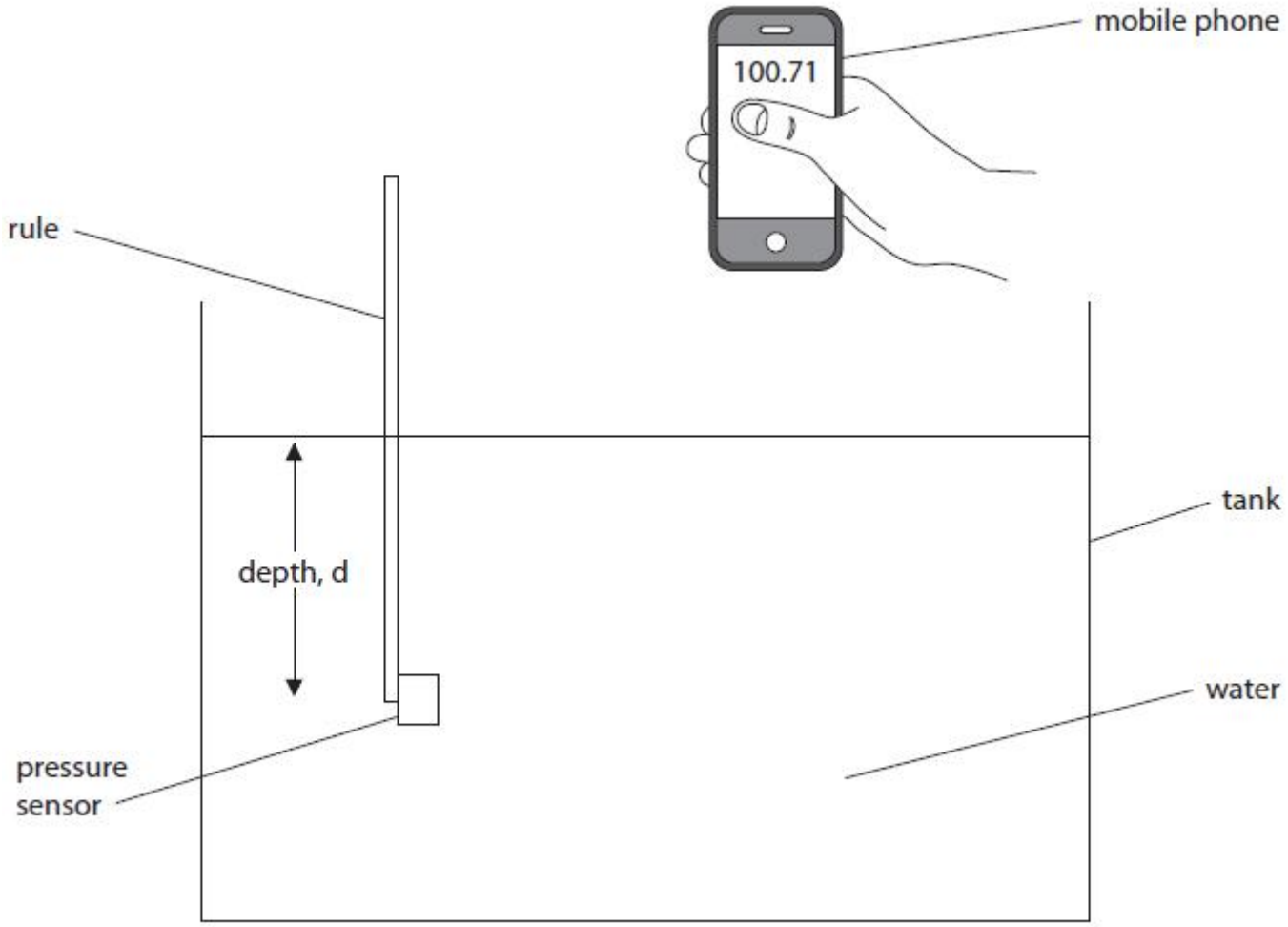

Figure 25

The depth of the pressure sensor below the surface of the water is read from the scale on the rule.

The pressure is displayed on the student's mobile phone which receives a signal from the pressure sensor.

Figure 26 gives some of the readings.

depth in m	pressure in kPa
0.050	99.15
0.100	99.70
0.150	100.15
0.200	100.70
0.250	101.15
0.300	101.70

Figure 26

Figure 27 shows a graph with some of the results plotted, but two of the points are missing.

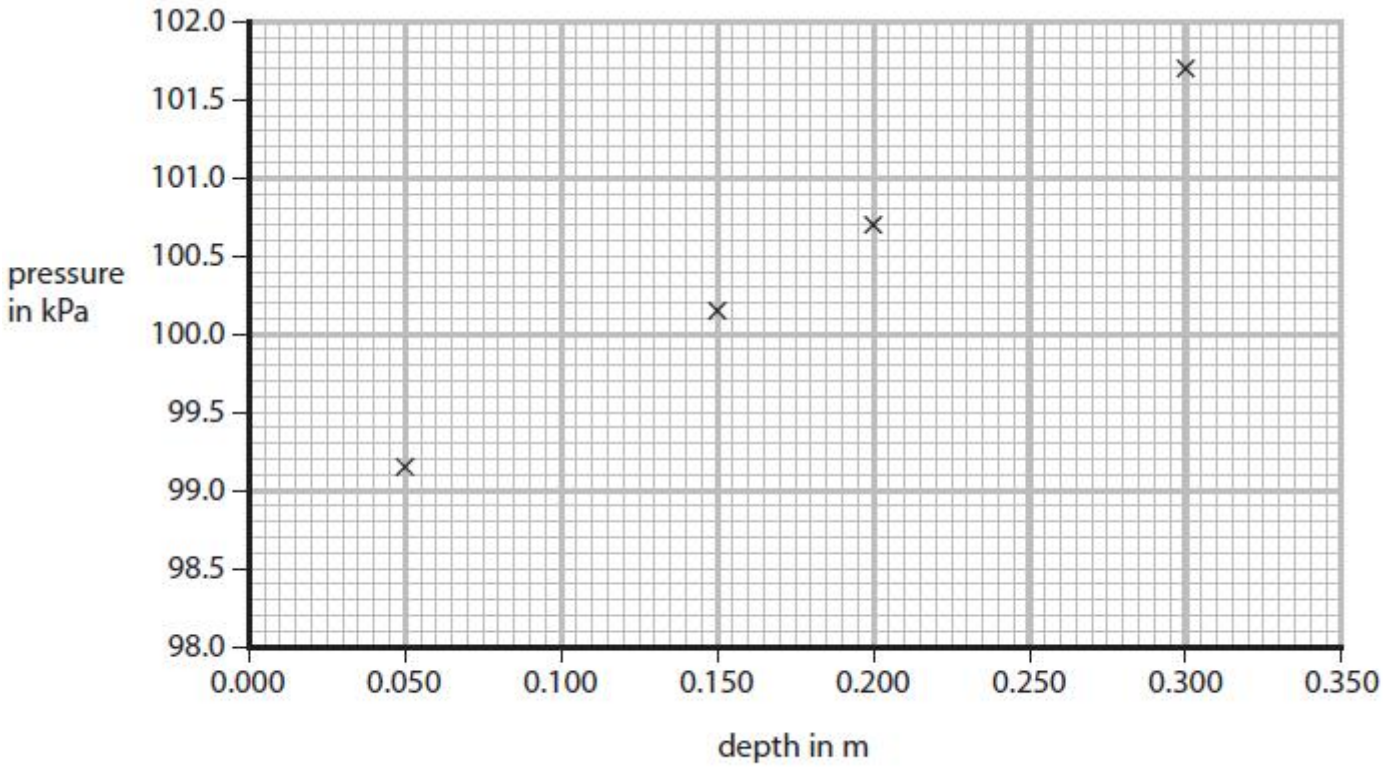

Figure 27

(i) Plot the two missing points on the graph.

(2)

(ii) Draw a line of best fit through the points on the graph.

(1)

(iii) Which of the following equations represents the variation of pressure with depth of water below the surface?

(1)

- ☐ **A** y = ax2 + b
- ☐ **B** y = mx
- ☐ **C** Y = mc - x
- ☐ **D** y = mx + c

(iv) Use the graph in Figure 27 to predict the pressure at the surface of the water.

(1)

pressure at the surface of the water = ... kPa

(Total for question = 5 marks)

Q4.

A student uses the apparatus shown in Figure 3 to investigate the extension of a spring.

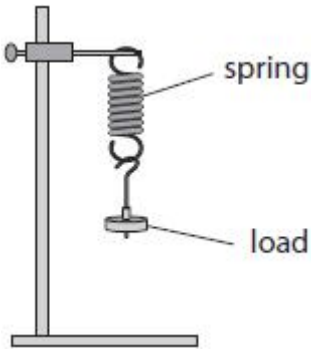

Figure 3

(i) Describe how the student could measure the extension of the spring when a load is added.

(3)

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(ii) The extension of the spring for a load of 1.5 N is 30 mm.

Calculate the spring constant for the spring.
Use the equation

$$\text{spring constant} = \frac{\text{load}}{\text{extension}}$$

(2)

spring constant = ... N/mm

(Total for question = 5 marks)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 16 shows a metal container with a movable piston.

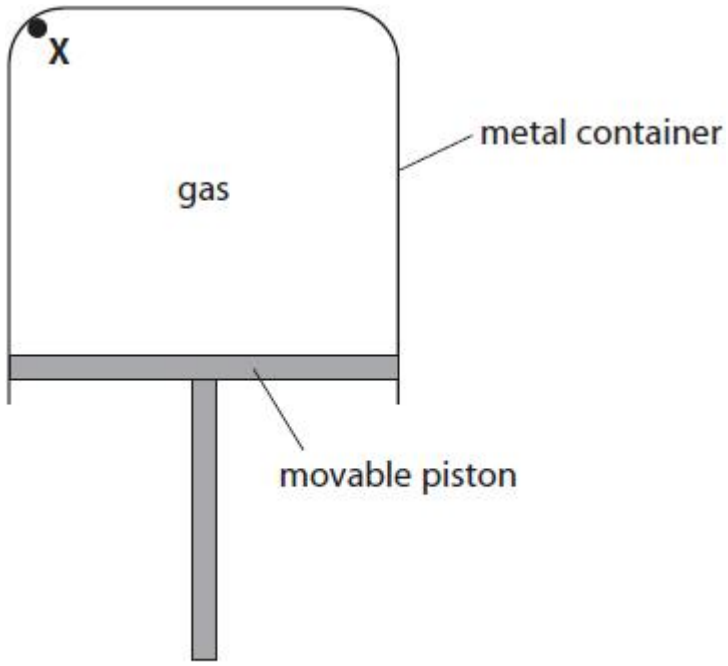


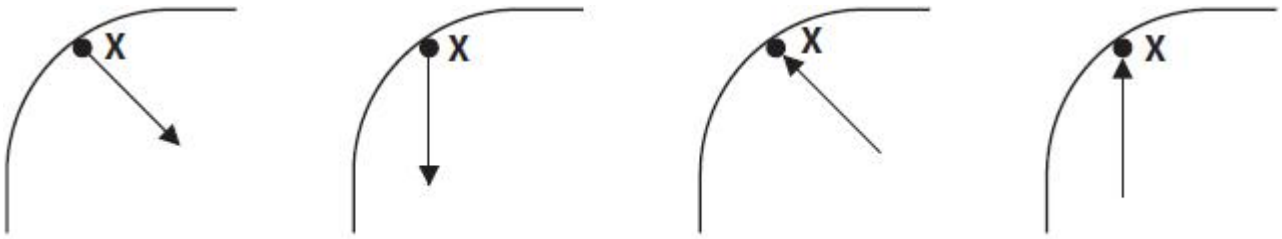
Figure 16

Point X is on the inner surface of the container.

The gas in the container is at a higher pressure than the air outside the container.

Which of these shows the direction of the force, due to the gas, on the container at point X?

(1)



☐ **A** ☐ **B** ☐ **C** ☐ **D**

(Total for question = 1 mark)

Q6.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 5 shows water coming out of holes in the side of a can.

The holes are at different heights.

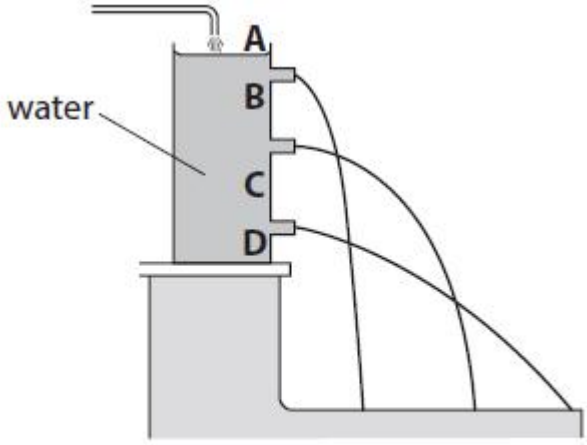


Figure 5

At which of these points is the pressure the lowest?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

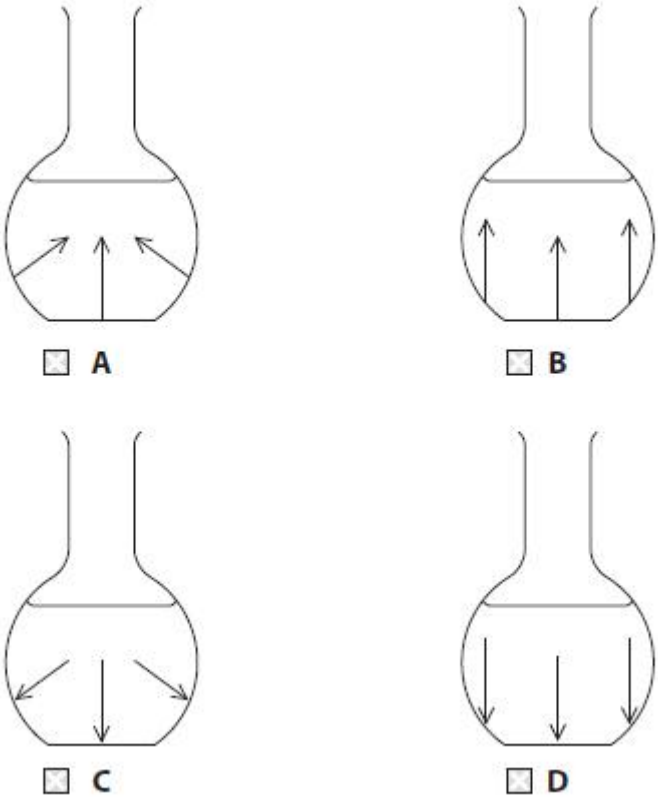
(Total for question = 1 mark)

Q7.

Water in a glass bottle exerts a pressure on the glass.

Which of these shows the correct directions of the pressure exerted by the water on the glass?

(1)



(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Bending or stretching objects can cause distortions.

Which of these small distortions can be described as **inelastic**?

(1)

- ☐ **A** stretching a rubber band
- ☐ **B** squashing a piece of modelling clay
- ☐ **C** compressing a steel spring
- ☐ **D** bending a plastic ruler

(Total for question = 1 mark)

Q9.

10 m of sea water exerts the same pressure as the atmosphere.

A submarine is at a depth of 50 m below the surface of the sea.

Calculate how many times greater the pressure is on the submarine than atmospheric pressure.

(2)

pressure = ... times greater

(Total for question = 2 marks)

Q10.

An aeroplane takes off from the ground.

State **two** factors that affect the pressure of the atmosphere on the aeroplane as the aeroplane goes higher in the atmosphere.

(2)

1 .....

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2 .....

.....

(Total for question = 2 marks)

Q11.

Figure 15 shows different water levels in two similar water containers with taps.

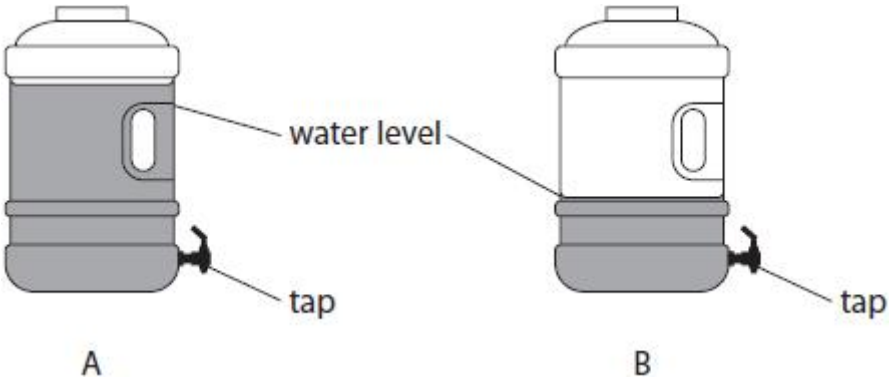

Figure 15

Explain why the water runs out of the tap of container A faster than out of the tap of container B.

(2)

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(Total for question = 2 marks)

Mark Scheme

Q1.

Question Number:	Answer	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative (example) content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. • same force at tip and head of the thumb tack • flat end has a large surface area • pointed end has a very small surface area • using pressure =force /area • at pointed end the pressure is large • large pressure , tip goes into wood • at flat end the pressure is much less • the flat end does not damage the thumb	**(1)** AO 1 2

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

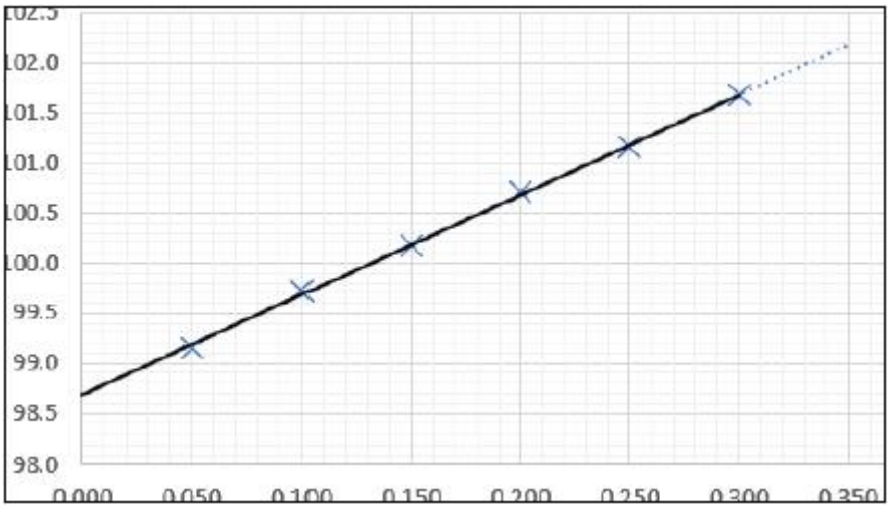
Q2.

Question number	Answer	Additional guidance	Mark
	An answer that contains the following points of understanding: • increase in depth (1) • increase in density (1)	sea water contains salt	(2)

Q3.

Question number	Answer	Additional guidance	Mark
(i)	Points plotted to within ± 1 small square (0.100, 99.7) (1) (0.250, 101.2) (1)		(2) AO2

Question number	Answer	Additional guidance	Mark
(ii)	best fit straight line passing through at least four of the points (1)	do not accept tramlining (multiple lines / curves) ignore slight shakiness in drawing	(1) AO2



Question number	Answer	Mark
(iii)	D $y = mx + c$ A is incorrect because the graph is a straight line and this equation describes a parabola. B is incorrect because the line intercepts the Y axis at a positive value and this equation describes a line passing through the origin. C is incorrect because this equation describes a line which intercepts the Y axis at a negative value.	(1) AO2

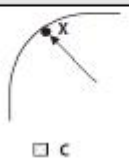
Question number	Answer	Additional guidance	Mark
(iv)	answer between 98.6 and 98.8 (kPa)	allow ecf from their line of best fit in b(ii)	(1) AO3

Q4.

Question number	Answer	Additional guidance	Mark
(i)	An answer that combines the following to provide a logical description of the method • measure unstretched length of spring (1) • measure stretched length of spring (1) • subtract (1)	 set unstretched position at 0 read stretched position use a ruler	(3)

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) $\frac{1.5}{30}$ evaluation (1) 0.05 (N/mm)	award full marks for correct answer without working 50 N/m allow power of 10 (POT) error for 1 mark	(2)

Q5.

Question number	Answer	Additional guidance	Mark
	 B and D are incorrect because they are not normal to the surface A is incorrect because the force should act outwards		(1)

Q6.

Question Number	Answer	Mark
	A the highest point in the can **B, C** and **D** are incorrect because they are lower in the can and more pressure is exerted	(1) AO1.1

Q7.

Question number	Answer	Mark
	C	**(1)**

Q8.

Question Number	Answer	Mark
	B squashing a piece of modelling clay **A** ,**C** and **D** are incorrect because all return to original shape when the force is removed. The distortion is elastic.	**(1)** **AO1.1**

Q9.

Question Number:	Answer	Additional guidance	Mark
	use of 50/10 (1) 6 (times)(1)	5 seen in answer line addition of 1 atmosphere award full marks for the correct answer without working	**(2)** AO 2 2

Q10.

Question number	Answer	Additional guidance	Mark
	Any two from: height of atmosphere (above aeroplane) (1) density of atmosphere (1) the temperature (of the atmosphere) (1)	 less air above the aeroplane accept oxygen for air in this context the air gets thinner the air gets colder	**(2)**

Q11.

Question Number:	Answer	Additional guidance	Mark
	an explanation linking 2 from: pressure in a liquid increases with depth (1) the greater the height of water (in the container)(1) the more force (pushing water out) (1)	 allow greater {weight of / volume of/amount of /more} water greater force/pressure/push (on water)	**(2)** AO 2 1